

Lab: Tables

Table Demo

Lightbulb Visions

You can view a common object many different ways.

For example, take a lightbulb. Whoever thought of visualizing a lightbulb exploding?



Lightbulb Glowing



A lightbulb drawing from Wikipedia



The last millisecond.
A photo by Cipaz CiCCiO
Flickr.com

Preview

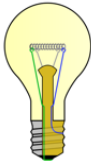
Tables allow you to organize row/column data on the page.

In this lab you will be setting up a basic table and then using the `colspan=""` and `rowspan=""` attributes to customize the table for headings and special text.

You'll also learn how to include images in a table and use CSS to style the table.

Here is what the finished page will look like:

Table Demo

Lightbulb Visions		
You can view a common object many different ways. For example, take a lightbulb. Whoever thought of visualizing a lightbulb exploding?	 Lightbulb Glowing	 A lightbulb drawing from Wikipedia
		 The last millisecond. A photo by Cipaz CiCCiO Flickr.com

Getting Setup

Begin with a basic HTML 5 web page. You'll want four rows and three columns.

Add this `<table>` code inside the `<body>` element of the page:

```
</head>  
<body>  
<h1>Table Demo</h1>
```

```
<table>  
  <!-- First row -->  
  <tr>  
    <td>r1c1</td>  
    <td>r1c2</td>  
    <td>r1c3</td>  
  </tr>  
  
  <!-- Second row -->  
  <tr>  
    <td>r2c1</td>  
    <td>r2c2</td>  
    <td>r2c3</td>  
  </tr>  
  
  <!-- Third row -->  
  <tr>  
    <td>r3c1</td>  
    <td>r3c2</td>  
    <td>r3c3</td>  
  </tr>  
  
  <!-- Fourth row -->  
  <tr>  
    <td>r4c1</td>  
    <td>r4c2</td>  
    <td>r4c3</td>  
  </tr>  
</table>  
</body>  
</html>
```

Add some CSS to add borders and set the width of each column:

```
<style type="text/css">
table {
  border:                1px solid black;
}

td {
  border:                1px solid black;
  width:                 15%;
}
</style>
```

Here is what the page should look like when viewed in a browser:

Table Demo

r1c1	r1c2	r1c3
r2c1	r2c2	r2c3
r3c1	r3c2	r3c3
r4c1	r4c2	r4c3

Here is all the code for the starting page:

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
<!-- tableDemo.html - Demonstration using
```

table elements

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peterk@webExplorations.com

Written: 04-04-14

Revised:

-->

<title>Table Demo</title>

<style type="text/css">

table {

border: 1px solid black;

}

td {

border: 1px solid black;

width: 15%;

}

</style>

</head>

<body>

<h1>Table Demo</h1>

<table>

```
<!-- First row -->
```

```
<tr>
```

```
    <td>r1c1</td>
```

```
    <td>r1c2</td>
```

```
    <td>r1c3</td>
```

```
</tr>
```

```
<!-- Second row -->
```

```
<tr>
```

```
    <td>r2c1</td>
```

```
    <td>r2c2</td>
```

```
    <td>r2c3</td>
```

```
</tr>
```

```
<!-- Third row -->
```

```
<tr>
```

```
    <td>r3c1</td>
```

```
    <td>r3c2</td>
```

```
    <td>r3c3</td>
```

```
</tr>
```

```
<!-- Fourth row -->
```

```
<tr>
```

```
    <td>r4c1</td>
```

```
        <td>r4c2</td>
        <td>r4c3</td>
    </tr>
</table>
</body>
</html>
```


Using colspan

If you want a title to cover several columns you can use the `colspan=" "` attribute.

```
<table>
  <!-- First row -->
  <tr>
    <td colspan="3">Lightbulb Visions</td>
    <td>r1c2</td>
    <td>r1c3</td>
  </tr>

  <!-- Second row -->
  <tr>
    <td>r2c1</td>
    <td>r2c2</td>
```

Uses 3 columns

These are now extra columns

But, notice that there are now five columns! The first `<td>` takes up three plus the other two `<td>` elements.

Here is what the page will look like. (Not quite what we were visualizing...)

Table Demo

Lightbulb Visions			r1c2	r1c3
r2c1	r2c2	r2c3		
r3c1	r3c2	r3c3		
r4c1	r4c2	r4c3		

Delete the two extra `<td>` elements. `r1c2` and `r1c3`.

Let's change the top row to `<th>` at the same time.

```
<table>
  <!-- First row -->
  <tr>
    <th colspan="3">Lightbulb Visions</th>
  </tr>

  <!-- Second row -->
  <tr>
    <td>1</td>
    <td>2</td>
    <td>3</td>
  </tr>
```

**Changed to:
table heading**

Don't forget to add a `th` selector up in the CSS `<style>` element.

```
<style type="text/css">
table {
  border: 1px solid black;
}

td {
  border: 1px solid black;
  width: 15%;
}

th {
  border: 1px solid black;
}
</style>
</head>
<body>
```

Add a selector for the `<th>`

Here is what the table should look like:

Table Demo

Lightbulb Visions		
r2c1	r2c2	r2c3
r3c1	r3c2	r3c3
r4c1	r4c2	r4c3

Using rowspan

You can use rowspan to create a vertical column that includes all the rows.

For example, lets say we want to have an explanation about all the photos.

In the second row <tr> add a new <td> like this:

```
<table>
  <!-- First row -->
  <tr>
    <th colspan="3">Lightbulb Visions</th>
  </tr>

  <!-- Second row -->
  <tr>
    <td>
      You can view a common object many different ways.
      For example, take a lightbulb.
      Whoever thought of visualizing a lightbulb exploding?
    </td>
    <td>r2c1</td>
    <td>r2c2</td>
    <td>r2c3</td>
  </tr>

  <!-- Third row -->
  <tr>
    <td>r3c1</td>
    <td>r3c2</td>
    <td>r3c3</td>
  </tr>

  <!-- Fourth row -->
  <tr>
    <td>r4c1</td>
```



Add another <td> in the second row.

The result will look like this. Notice how the new column pushes the existing information off the right side of the table.

Table Demo

The new `<td>` pushes things off the right edge of the table

Lightbulb Visions			
You can view a common object many different ways. For example, take a lightbulb. Whoever thought of visualizing a lightbulb exploding?	r2c1	r2c2	r2c3
r3c1	r3c2	r3c3	
r4c1	r4c2	r4c3	

Add a `rowspan=" "` attribute to the `<td>` to make the cell "grow" downward. This will align all the columns with a deep column, spanning the three rows of data.

Here is the code:

```
<table>
  <!-- First row -->
  <tr>
    <th colspan="3">Lightbulb Visions</th>
  </tr>

  <!-- Second row -->
  <tr>
    <td rowspan="3">
      You can view a common object many different ways.
      For example, take a lightbulb.
      Whoever thought of visualizing a lightbulb exploding?
    </td>
    <td>r2c1</td>
    <td>r2c2</td>
    <td>r2c3</td>
  </tr>
```

rowspan causes the cell to "grow" downward.

And here is how the table should look:

Table Demo

Lightbulb Visions			
You can view a common object many different ways. For example, take a lightbulb. Whoever thought of visualizing a lightbulb exploding?	r2c1	r2c2	r2c3
	r3c1	r3c2	r3c3
	r4c1	r4c2	r4c3

The rowspan="3" tells the browser to "grow" the cell down three rows.

It also adds a new column, leaving the <th> short. Increase the colspan=" " to accommodate the new col.

Finally adjust the colspan=" " attribute on the heading so it spans four columns. The original colspan="3" didn't take the additional description column into account.

Here is the code:

```

<table>
  <!-- First row -->
  <tr>
    <th colspan="4">Lightbulb Visions</th>
  </tr>

  <!-- Second row -->
  <tr>
    <td rowspan="3">
      You can view a common object many different ways.
      For example, take a lightbulb.
      Whoever thought of visualizing a lightbulb exploding?
    </td>
    <td>r2c1</td>
    <td>r2c2</td>
    <td>r2c3</td>
  </tr>

  <!-- Third row -->
  <tr>
    <td>r3c1</td>

```

One <td> spanning four columns

**Four separate columns.
The first <td> goes down three rows**

Here is what the output will look like with all the columns all lined up.

Table Demo

4 columns

Lightbulb Visions			
You can view a common object many different ways. For example, take a lightbulb. Whoever thought of visualizing a lightbulb exploding?	r2c1	r2c2	r2c3
	r3c1	r3c2	r3c3
	r4c1	r4c2	r4c3

3 rows

Add Images and Captions

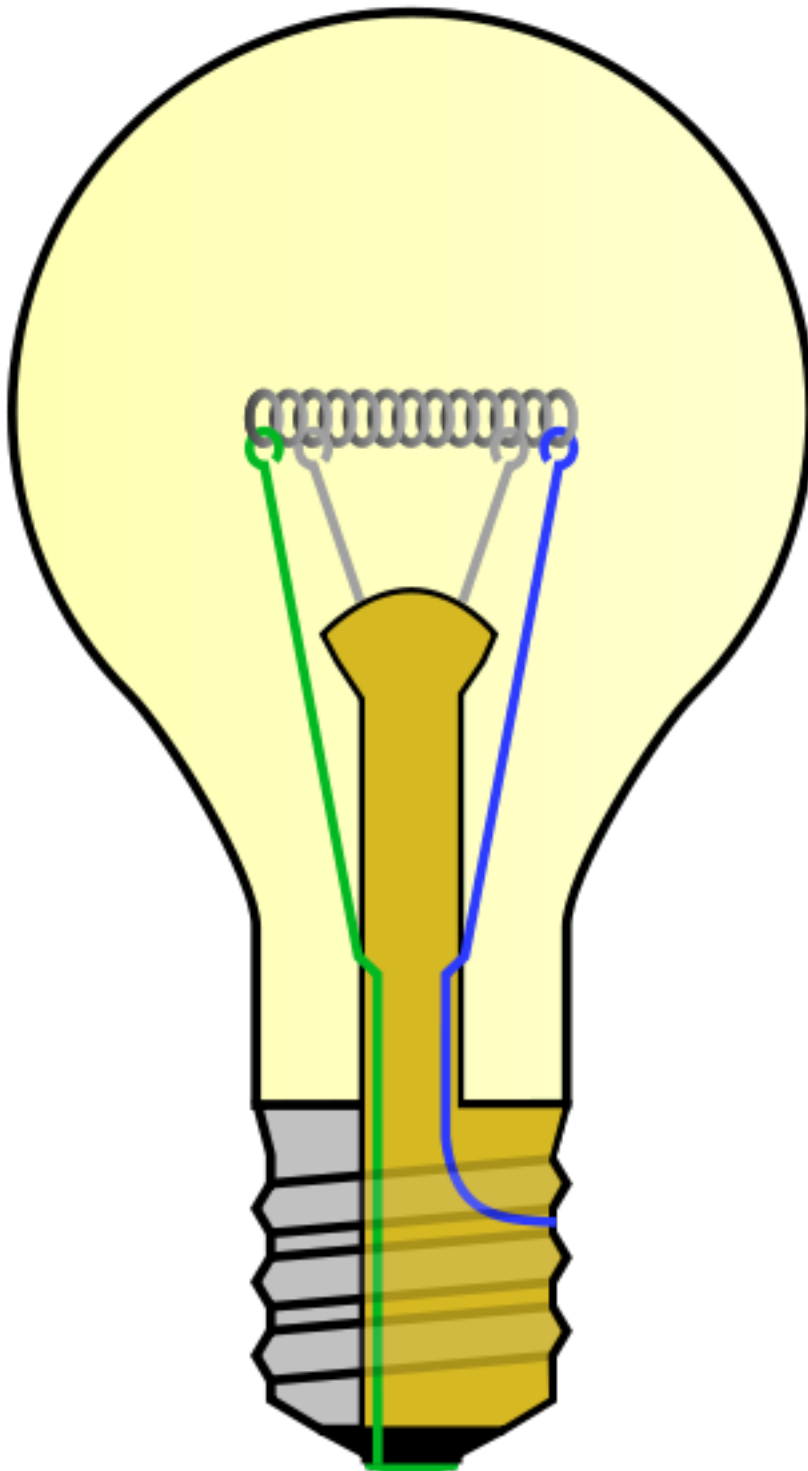
Here are three images that you can add to the table. Download them by right-mouse clicking on the image and selecting "Save image as"



lightbulb.png



lightbulbBurnout.png



lightbulbDrawing.png

To add these to the table replace the text with an `<img />` element.

Here is the code:

```
<table>
  <!-- First row -->
  <tr>
    <th colspan="4">Lightbulb Visions</th>
  </tr>

  <!-- Second row -->
  <tr>
    <td rowspan="3">
      You can view a common object many different ways.
      For example, take a lightbulb.
      Whoever thought of visualizing a lightbulb exploding?
    </td>
    <td>
      
    </td>
    <td>
      
    </td>
    <td>
      
    </td>
  </tr>

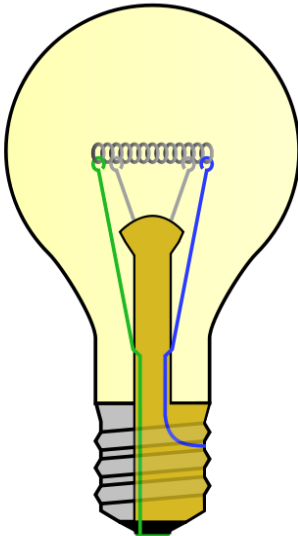
  <!-- Third row -->
  <tr>
    <td>r3c1</td>
    <td>r3c2</td>
    <td>r3c3</td>
  </tr>

  <!-- Fourth row -->
  <tr>
    <td>r4c1</td>
```

Adding images to a table.



Table Demo

Lightbulb Visions			
You can view a common object many different ways. For example, take a lightbulb. Whoever thought of visualizing a lightbulb exploding?			
r3c1		r3c2	r3c3
r4c1		r4c2	r4c3

Notice how the largest object determines how high and how wide the cells are.

In the next row add in some captions.

It looks like we have an extra row so let's delete that last row and adjust the rowspan=" " attribute.

```

<table>
  <!-- First row -->
  <tr>
    <th colspan="4">Lightbulb Visions</th>
  </tr>

  <!-- Second row -->
  <tr>
    <td rowspan="2">
      You can view a common object many different ways.
      <br /><br /><br />
      For example, take a lightbulb.
      Whoever thought of visualizing a lightbulb exploding?
    </td>
    <td class="image">
      
    </td>
    <td class="image">
      
    </td>
    <td class="image">
      
    </td>
  </tr>

  <!-- Third row -->
  <tr>
    <td>Lightbulb Glowing</td>
    <td>A lightbulb drawing from Wikipedia</td>
    <td>
      The last millisecond.<br />
      A photo by Cipaz CiCCi0<br />
      <a href="http://flickr.com" target="_blank">Flickr.com</a>
    </td>
    <td></td>
  </tr>
</table>

```

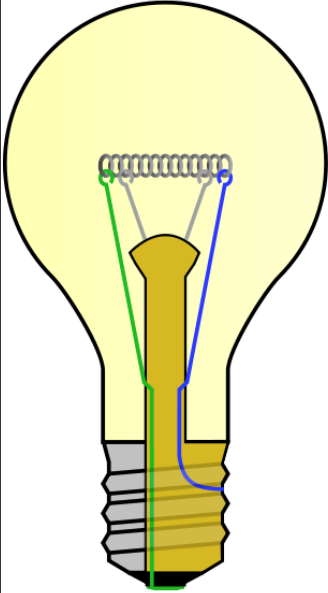
Adjust the rowspan="" to accommodate deleting the fourth row.

Add captions in the third row

Delete the fourth row. Guess we didn't need it ;-)

Here is what the table should look like:

Table Demo

Lightbulb Visions			
You can view a common object many different ways. For example, take a lightbulb. Whoever thought of visualizing a lightbulb exploding?			
	Lightbulb Glowing	A lightbulb drawing from Wikipedia	The last millisecond. A photo by Cipaz CiCCiO Flickr.com

CSS and Tables

Use CSS to change the appearance of the table.

You can change the size of the images working with the img selector.

You can change the alignment of various cells by using a class to designate each type of cell.

For each <td> holding a graphic add a class="graphic" attribute.

For each <td> holding a caption add a class="caption" attribute.

Here is the code:

```
<table>
  <!-- First row - Heading -->
  <tr>
    <th colspan="4">Lightbulb Visions</th>
  </tr>

  <!-- Second row - Graphic Images -->
  <tr>
    <td rowspan="2">
      You can view a common object many different ways.
      <br /><br /><br />
      For example, take a lightbulb.
      Whoever thought of visualizing a lightbulb exploding?
    </td>
    <td class="graphic">
      
    </td>
    <td class="graphic">
      
    </td>
    <td class="graphic">
      
    </td>
  </tr>

  <!-- Third row - Captions -->
  <tr>
    <td class="caption">Lightbulb Glowing</td>
    <td class="caption">A lightbulb drawing from Wikipedia</td>
    <td class="caption">
      The last millisecond.<br />
      A photo by Cipaz CiCCiO<br />
      <a href="http://flickr.com" target="_blank">Flickr.com</a>
    </td>
  </tr>
</table>
```

In the `<style>` section of the page add these selectors and rules.

For a cleaner appearance you can change the font-family for the whole `<table>` to sans-serif type of fonts.

Make the font-size twice as large for the `<th>` heading.

Add the `.graphic` and `.caption` class selectors.

Change the height so all the images in the table are the same height.

Here is the CSS code:


```
<style type="text/css">
table {
border: 1px solid black;
font-family: helvetica, arial, sans-serif;
}

td, th {
padding: 10px;
}

td {
border: 1px solid black;
width: 7%;
}

th {
border: 1px solid black;
font-size: 2em;
}

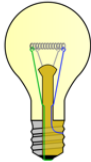
td.graphic {
text-align: center;
vertical-align: bottom;
}

td.caption {
text-align: center;
vertical-align: top;
}

td img {
height: 150px;
}
</style>
```

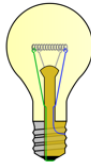
This is what the page should look like:

Table Demo

Lightbulb Visions			
You can view a common object many different ways. For example, take a lightbulb. Whoever thought of visualizing a lightbulb exploding?			
	Lightbulb Glowing	A lightbulb drawing from Wikipedia	The last millisecond. A photo by Cipaz CiCCiO Flickr.com

Finally, remove all the borders except for the <table> border displaying a modern-looking page:

Table Demo

Lightbulb Visions			
You can view a common object many different ways. For example, take a lightbulb. Whoever thought of visualizing a lightbulb exploding?			
	Lightbulb Glowing	A lightbulb drawing from Wikipedia	The last millisecond. A photo by Cipaz CiCCiO Flickr.com

Complete Source Code

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <!-- tableDemo.html - Demonstration using
table elements
  Peter Johnson -
peterk@webExplorations.com
  Written:    04-04-14
  Revised:
  -->
  <title>Table Demo</title>

  <style type="text/css">
table {
border:                                1px solid black;
font-family:                          helvetica, arial,
sans-serif;
}

td, th {
padding:                                10px;
}
```

```
td {  
  /* border:                1px solid  
black; */  
  width:                    7%;  
}
```

```
th {  
  /* border:                1px solid  
black; */  
  font-size:                2em;  
}
```

```
td.graphic {  
  text-align:               center;  
  vertical-align:           bottom;  
}
```

```
td.caption {  
  text-align:               center;  
  vertical-align:           top;  
}
```

```
td img {
```

```

height:                                150px;
}
</style>

</head>
<body>
<h1>Table Demo</h1>

<table>
  <!-- First row - Heading -->
  <tr>
    <th colspan="4">Lightbulb Visions</
th>
  </tr>

  <!-- Second row - Graphic Images -->
  <tr>
    <td rowspan="2">
      You can view a common object many
different ways.
      <br /><br /><br />
      For example, take a lightbulb.
      Whoever thought of visualizing a
lightbulb exploding?

```

```

        </td>
        <td class="graphic">
            
        </td>
        <td class="graphic">
            
        </td>
        <td class="graphic">
            
        </td>
    </tr>

    <!-- Third row - Captions -->
    <tr>
        <td class="caption">Lightbulb
Glowing</td>
        <td class="caption">A lightbulb
drawing from Wikipedia</td>
        <td class="caption">

```

```
        The last millisecond.<br />
        A photo by Cipaz CiCCiO<br />
        <a href="http://flickr.com"
target="_blank">Flickr.com</a>
    </td>
</tr>
</table>

</body>
</html>
```

Credits

Lab: Tables by Peter K. Johnson - WebExplorations.com
April 2014



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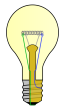
Photo Credits



<http://Flickr.com> Photo by TschAe



<http://Flickr.com> Photo by Cipaz CiCCiO



[http://commons.wikimedia.org/wiki/File:Incandescent_light_bulb_\(no_labels\).svg](http://commons.wikimedia.org/wiki/File:Incandescent_light_bulb_(no_labels).svg)